CONFIDENTIAL

OralSentinel

Oral lesion screening from a single clinical photograph

Research report: results, benchmark against published work, and candidate inventions

4 November 2026

Research prototype. Not a medical device and not for diagnostic use.

Confidentiality. Section 8 describes candidate inventions that have not been filed. Share this document only under a signed non-disclosure agreement (NDA). Publishing or presenting it before a patent filing could prevent the inventions from being patented.

1. Executive summary

OralSentinel analyses **one photograph** of an oral lesion with the patient's details and returns a dysplasia grade, a cancer result, a heatmap and a PDF report. A second photo of the patient's healthy opposite side is optional.

- **Screening result on held-out data.** A new risk-controlled triage layer (cancer / inconclusive / no cancer) flags or refers **98.3%** of cancers (95% CI 90.8-100%) while correctly clearing **81.8%** of non-cancers, from one photo. Only 1 of 58 test cancers (1.7%) was called "no cancer", inside the 5% bound the method promises.
- **Against published work.** The most comparable published external test (Fu et al. 2020, 44,409 training photos) reports 89.6% sensitivity and 80.6% specificity. OralSentinel is higher on both; even the lower end of its sensitivity interval (90.8%) is above 89.6%. The photos differ, so this is indicative, not head-to-head.
- **Single-answer model.** AUC 0.932 (95% CI 0.887-0.971), sensitivity 84.5%, specificity 87.0% on the corrected test set.
- **Research finding.** The public Kaggle dataset used by many papers files 72 benign lesions as cancer, and common evaluation shortcuts add about 5 accuracy points. Published 94-99% figures on such data are inflated; OralSentinel is evaluated without these shortcuts.
- **Weaknesses.** Close-ups of a normal inner cheek and benign lichen planus are still often flagged; the dysplasia grade is right for only about 40-45% of patients.

Verdict: a working research platform whose screening numbers are competitive with published external results, with honest, documented limits. Not yet clinically validated.

2. What the product does

Step	What happens
Sign in	Secure accounts; each clinician sees only their own patients
Upload	One lesion photo (required), optional opposite-side photo, patient details (age, sex, tobacco and betel quid use with years, alcohol, lesion site and side, habit side, symptoms, history, notes)
Photo check	Non-mouth images (slides, screenshots, blank images) are refused and nothing is stored
Result	Dysplasia grade (none / mild / moderate / severe) and cancer result (cancer / inconclusive / no cancer), shown against all possible answers; no raw scores
Explainability	Heatmap overlay showing where the model looked
Report and history	Downloadable PDF; dashboard and history of past analyses

Automated checks: 43 backend tests pass; the frontend type-checks, lints and builds; the workflow was checked in a browser against the running models.

3. Data

Source	Content	Use
Kaggle oral cancer dataset (v2.0)	950 photos; 902 distinct after grouping duplicates	Training, validation, test
Client case sheet and folders	196 graded photos from 99 patients (24 mild, 39 moderate, 36 severe dysplasia) + 1 carcinoma	Training, held-out client test, grading
Mendeley "Oral Images Dataset" (CC BY 4.0)	165 benign, 158 malignant photos, Karnataka hospitals	Label cross-check
Mendeley lichen planus example set (CC BY 4.0)	5 healthy cheek + 5 lichen planus photos	External test only

Patient names and identifiers are never copied into the code repository. Every split is by patient, so one patient never appears in both training and testing.

4. Results: every metric, with 95% confidence intervals

Model in use: round 7 (ResNet-50, trained on corrected labels with shortcut-removal augmentation). Validation photos were used to choose models and thresholds; test photos were used once. Confidence intervals: 2,000 bootstrap resamples.

Single answer (threshold 0.5)

Metric	Validation (175 photos, 62 cancers)	Test (135 photos, 58 cancers)
Accuracy	89.7% (85.1-94.3)	85.9% (80.0-91.1)
Sensitivity (recall)	87.1% (78.3-94.7)	84.5% (74.5-93.3)
Specificity	91.2% (85.8-96.3)	87.0% (79.2-94.0)
Precision (PPV)	84.4% (74.6-92.9)	83.1% (73.7-91.8)
NPV	92.8% (87.8-97.3)	88.2% (80.8-94.8)
F1 score	0.857 (0.782-0.917)	0.838 (0.762-0.902)
Balanced accuracy	89.1% (83.7-93.9)	85.7% (79.7-91.2)
Matthews correlation (MCC)	0.777 (0.671-0.869)	0.714 (0.595-0.822)
Cohen's kappa	0.777 (0.669-0.869)	0.713 (0.594-0.821)
ROC-AUC	0.943 (0.909-0.973)	0.932 (0.887-0.971)
PR-AUC	0.900 (0.832-0.955)	0.883 (0.794-0.958)
Brier score (lower is better)	0.091 (0.059-0.125)	0.112 (0.070-0.155)
Expected calibration error	0.074 (0.052-0.129)	0.094 (0.064-0.159)

Client held-out patients (39 photos, all dysplasia, no cancer): specificity 87.2%.

Risk-controlled triage (single photo, in use)

The photo is re-analysed under nine changes that should not alter the diagnosis; the average cancer probability decides. The "no cancer" threshold is set on validation by conformal risk control so that at most 5% of cancers are expected to be called "no cancer"; the "cancer" threshold keeps non-cancers called cancer near 10%.

Test set (58 cancers, 77 non-cancers)	Plain threshold	Vote triage (earlier)	Risk-controlled triage
Cancers called "no cancer"	15.5%	3.4%	1.7% (95% CI 0.0-9.2)
Cancers flagged or referred	84.5%	96.6%	98.3% (90.8-100)
Non-cancers cleared	87.0%	84.4%	81.8% (71.4-89.7)
Non-cancers called "cancer"	13.0%	13.0%	13.0% (6.4-22.6)
Inconclusive (refer)	-	8.1%	8.9%

The promised bound held on unseen data (1.7% against 5%). A stricter version that holds the bound with 90% confidence missed no test cancers but sent 30% of photos to "inconclusive", too many with only 62 calibration cancers. Many client dysplasia photos come out inconclusive; these patients have real lesions, so referral is clinically reasonable.

5. Benchmark against published work

Study	Data	Evaluation	Sensitivity	Specificity	AUC
Fu et al. 2020, EClinicalMedicine	44,409 photos, 11 hospitals	Internal	94.9%	88.7%	0.983

Study	Data	Evaluation	Sensitivity	Specificity	AUC
Fu et al. 2020	same model	External (journal photos)	89.6%	80.6%	0.935
Warin et al. 2021 (DenseNet121)	700 photos, cancer vs normal	Internal	98.75%	100%	0.99
Jubair et al. (EfficientNet-B0)	Clinical images	Internal	-	-	0.928 (acc. 85%)
Meta-analysis, 8 photo studies	Pooled	Mixed	97%	93%	0.99
Review, 14 clinical-image studies	Pooled	Mixed	86%	67%	-
Frontiers 2024 meta-analysis	Pooled, all imaging	Mixed	87%	81%	-
OralSentinel single answer	Corrected, deduplicated	Held-out test	84.5%	87.0%	0.932
OralSentinel risk-controlled triage (refer if cancer or inconclusive)	Corrected, deduplicated	Held-out test	98.3%	81.8%	-

Different studies use different data and protocols; only results on the same test set are directly comparable. Internal results tend to be higher than external ones (Fu: AUC 0.983 internal vs 0.935 external).

Why published figures are often inflated

The same simple classifier (frozen ImageNet ResNet-50 features + logistic regression) was evaluated on the public Kaggle dataset under four protocols, 5 random splits each:

Evaluation protocol	Accuracy	AUC
Augmented copies made before the train/test split; original labels	90.9%	0.970
Random split, duplicates kept; original labels	85.8%	0.943
Duplicates kept on one side of the split; original labels	89.7%	0.960
Duplicates grouped and labels corrected (honest)	85.6%	0.931

About 5 accuracy points and 0.04 AUC come from the protocol and label errors, not the model. Duplicates had little effect here (only 48 photos); the gap between the two middle rows mostly reflects which photos fall in the test set. The label errors matter most: 72 photos the original Mendeley source files as benign lesions sit in Kaggle's cancer folder (about 1 in 6 of the "cancers" used in training), which turns part of the task into "lesion vs no lesion" and makes it look easier than real cancer detection. A 2026 study also reports augmentation leakage in oral cancer image work; the label errors were not found reported elsewhere.

6. Development and robustness

Hidden shortcuts found and removed

Zooming into healthy photos raised false cancer calls from 1.5% to 20% (the public healthy photos are mostly whole-mouth phone pictures, its cancers mostly clinical close-ups), and ordinary photo changes made the model miss cancers: zoomed out 69% caught, glare 73%, heavy JPEG 74%, low resolution or white border 76%, blur 77% (normally 84%). Training with these changes applied to both classes cut false calls on zoomed healthy photos to 6.5%.

Photo check

Refuses non-mouth images: histopathology slides 95%, wallpapers 100%, blank / noise / document images 100%; 0 of 174 held-out mouth photos wrongly refused. Faces, skin and teeth were not available to test.

Input robustness

400 random and nonsense patient records found four bugs (a crash on nested input, contradictory habit answers stored, durations longer than the patient's age, 1/0 accepted as yes/no); all fixed, with no crashes and nothing contradictory stored afterwards.

7. Dysplasia grading

	Photo model (ImageNet start)	Photo model (cancer-model start)	Always "moderate"
Accuracy (photos, 5-fold, patient-grouped)	44.9%	40.8%	39.3%
Weighted kappa	0.25 (fair)	0.06	0

Patient details alone reached kappa 0.15, which shuffled-grade runs matched 13% of the time (chance level); photo plus details did not help. A "no dysplasia" answer learned from healthy mouths is given to 98.5% of healthy test mouths and to 1 of 39 client dysplasia photos. Shown with "Low reliability: matched the biopsy grade for about 40% of patients in testing".

8. Candidate inventions (CONFIDENTIAL)

Not filed. Do not publish, present or share outside an NDA before a patent attorney files.

Invention 1 (single photo, validated): risk-controlled acquisition-consensus triage

Three layers on top of any image classifier: (1) an **acquisition audit** measures which capture factors (zoom, glare, compression, framing, rotation, exposure) change the model's answer; (2) the photo is re-analysed under a fixed set of such diagnosis-preserving changes and the **mean cancer probability** is the consensus score; (3) **conformal risk control** on a calibration set fixes the "no cancer" threshold so the expected share of cancers called "no cancer" is at most a chosen level (5%), and a second threshold bounds false cancer calls; between them the result is **inconclusive: retake or refer**.

Evidence (corrected held-out test): 1.7% of cancers called "no cancer" (bound 5%); 98.3% flagged or referred (95% CI 90.8-100%) with 81.8% of non-cancers cleared, against 84.5% / 87.0% for a plain threshold, and above the best comparable published external result (Fu et al. 2020: 89.6% / 80.6%).

Prior art to address: test-time augmentation for uncertainty (Ayhan and Berens 2018; Wang et al. 2019), conformal prediction and risk control (Angelopoulos et al. 2022) and its use in medical imaging. The claimed contribution is the combination: an acquisition-audit-derived edit set, a consensus score, and a risk-controlled referral output with a stated bound on missed cancers for oral screening. Novelty: moderate; strongest evidence.

Disclosure: the owner chose to deploy the full version publicly; once live, this method is in public use and likely no longer patentable in India or Europe. File before going live if patent protection is wanted.

Invention 2 (candidate): same-patient self-reference

The lesion is compared with the same patient's own healthy tissue rather than with other people's mouths, cancelling camera, light, pigmentation and tobacco staining. It works in three forms, from one photo upward: (a) the most suspicious region against the surrounding tissue in the same photo, located by the model's activation map; (b) left half against mirrored right half of a whole-mouth photo; (c) an optional photo of the matching opposite-side site. Safety rules: the reference can only move a result towards "inconclusive", never to "no cancer"; an abnormal-looking reference triggers "examine both sides".

Evidence	Result
(b) left/right asymmetry alone, cancer vs non-cancer (AUC, validation / test)	0.68 / 0.76
(a) lesion-vs-surround contrast alone (test AUC)	0.09, i.e. 0.91 in reverse: cancer photos show less contrast, likely because the lesion fills the frame
(a)+(b) combined with the model (test AUC)	0.930 / 0.928 vs 0.932: no overall gain yet
(a) at a high catch rate (test)	sensitivity 96.6% vs 94.8%, specificity 81.8% vs 79.2%, client dysplasia flagged 41% vs 54%: promising, within noise
(c) different-person reference	no separation; a true same-patient pair is required

Prior art to address: contralateral sites imaged as normal samples (You et al. 2026); contralateral normal sites in optical spectroscopy (US 6,571,118; WO2015077684A1); intra-patient comparison in dermatology ("ugly duckling", Soenksen et al. 2021). Novelty: plausible as a specific combination, but an examiner may argue obviousness; proof of added value needs paired same-patient photos.

Methods tested that did not help

Method	Result on held-out data
Pathology-ordered model (normal < benign < mild < moderate < severe < cancer, one ordinal network)	Validation AUC 0.901 vs 0.943; client dysplasia flagged 54% vs 13%; grade kappa 0.07. Worse; not used.
Lesion-vs-surround comparison in the same photo, added to the model	Test AUC 0.930 vs 0.932; no overall gain
Colour correction using a reference photo	Accuracy fell from 87% to 74-79%; dropped

Research contributions (publish only after filing)

- Label errors in a widely used public oral cancer dataset (72 benign lesions filed as cancer) and their effect on reported accuracy.
- Quantified inflation from evaluation shortcuts (about 5 accuracy points, 0.04 AUC).
- An acquisition-shortcut audit for oral photo models (zoom, glare, compression, framing) and its correction.

Assessment

Invention 1 has the strongest evidence and works on one photo; Invention 2 addresses the main failure (normal tissue flagged) but is unproven. In India software alone is excluded (Patents Act s.3(k)), so claims should be framed as a capture-and-analysis system with a technical effect (robustness to acquisition conditions). Recommendation: a professional prior-art search and patent attorney opinion before filing; keep all of the above private until then.

Disclosure status

Already public (cannot be claimed): the photo check, shortcut augmentation, label correction, and grading model, pushed earlier to a public code repository. Not public: Inventions 1 and 2, their experiments and this report, held only in a private repository.

9. Limitations

- **Small test sets:** 58 test cancers; one cancer is 1.7%, so intervals are wide.
- **Client cancer data:** one confirmed cancer, so detection on the client's own photos cannot be measured.
- **Normal cheek close-ups and benign lesions:** 4 of 5 external healthy cheeks and 4 of 5 lichen planus photos flagged.
- **Grading:** right for about 40-45% of patients.
- **Calibration:** expected calibration error 0.09 on test; probabilities are not shown to users.
- **No clinical validation, ethics approval or regulatory pathway yet; not cloud-deployed yet.**

10. Next steps

1. **Patent:** keep the repository private; professional prior-art search; patent attorney; consider an Indian provisional application; NDA for any sharing; trademark search for "OralSentinel".
2. **Pilot with the client:** confirmed cancers and benign lesions from their clinic, normal cheek close-ups, and, where possible, photos of both sides of each patient (150-300 patients). This measures cancer detection on their photos and tests Invention 2.
3. **Clinical decisions:** confirm the three-way rule and how "inconclusive" is acted on.
4. **Deployment:** secure cloud deployment for the pilot (encrypted storage, backups, monitoring), behind the research-prototype disclaimer.
5. **Publication:** after filing, write up the benchmark findings (section 5) as a paper.

References

Fu Q. et al. A deep learning algorithm for detection of oral cavity squamous cell carcinoma from photographic images: a retrospective study. EClinicalMedicine 27:100558, 2020.

Warin K. et al. Automatic classification and detection of oral cancer in photographic images using deep learning algorithms. 2021.

Jubair F. et al. A novel lightweight deep convolutional neural network for early detection of oral cancer. Oral Diseases, 2022.

Nieri M. et al. Diagnosis of oral cancer with deep learning: a comparative test accuracy systematic review. Oral Diseases, 2025.

Artificial intelligence for oral cancer detection: systematic review and meta-analysis. Frontiers in Oral Health, 2024. doi:10.3389/froh.2024.1494867.

Accuracy of image-based deep learning in oral squamous cell carcinoma diagnosis: systematic review and meta-analysis. PubMed 42556899.

Data leakage concerns in training and evaluation protocols for oral cancer image classification. SN Computer Science, 2026. doi:10.1007/s42979-026-05141-y.

Talwar V. et al. AI-assisted screening of oral potentially malignant disorders using smartphone-based photographic images. Cancers 15:4120, 2023.

Birur N.P. et al. Role of community health worker in a mobile health program for early detection of oral cancer. 2019.

You R. et al. Explainable multimodal deep learning integrating imaging and clinical data for OPMD detection. arXiv:2609.04512, 2026.

US Patent 6,571,118. Combined fluorescence and reflectance spectroscopy. WO2015077684A1. Colposcopes having light emitters and image capture devices.

Soenksen L.R. et al. Using deep learning for dermatologist-level detection of suspicious pigmented skin lesions from wide-field images. Sci. Transl. Med., 2021.

Angelopoulos A.N. et al. Conformal risk control. arXiv:2208.02814, 2022.

Ayhan M.S., Berens P. Test-time data augmentation for estimation of heteroscedastic aleatoric uncertainty in deep neural networks. MIDL, 2018.

Wang G. et al. Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation. Neurocomputing, 2019.

Chandrashekar H.S. et al. Oral Images Dataset. Mendeley Data, doi:10.17632/mhjyrn35p4.2 (CC BY 4.0).

Theppitak S. et al. Photorealistic synthesis of oral lichen planus lesions: code and example dataset. Mendeley Data fgjd95tcr8 (CC BY 4.0).